

HW6

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Q6

Inputs

- `pdb_id`: PDB ID or file name
- `chain`: chain to analyze (default = "A")
- `elety`: atom type to analyze (default = "CA")

What the function does and how to use it

This function reads in protein PDB data, trims it to the specified chain and atom type, extracts the B-factors, and outputs a plot for the specified protein.

Use it by calling `plot_bfactor("4AKE")` or another PDB ID.

Output

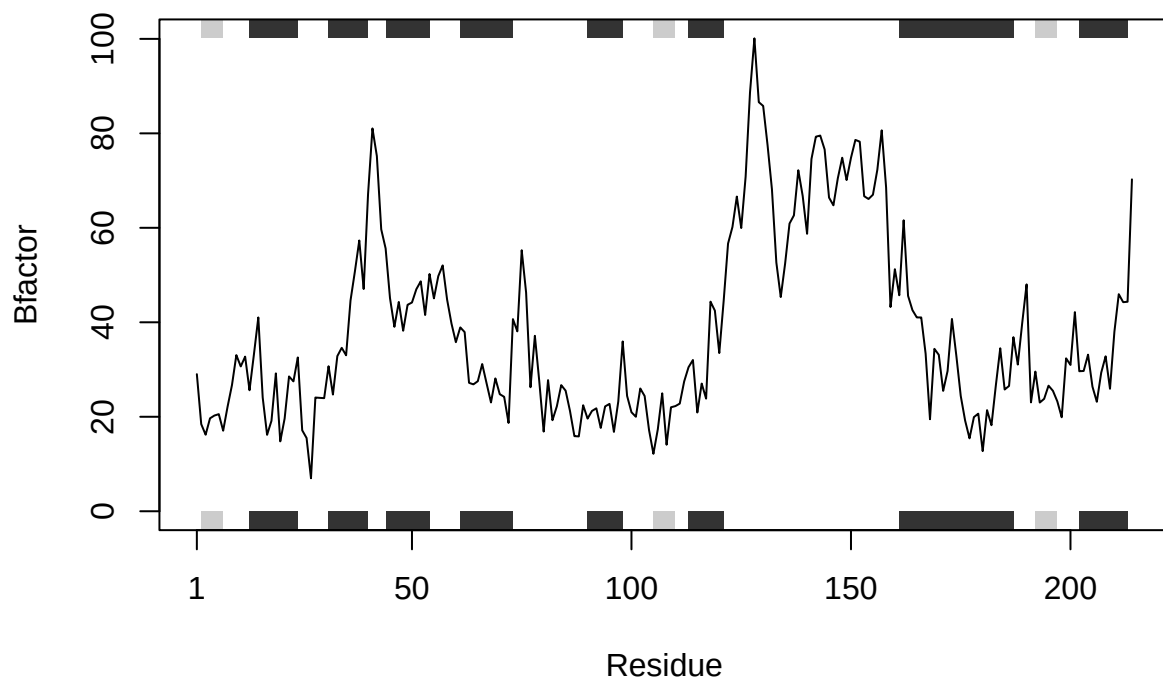
The function outputs a B-factor plot for the specified protein and returns a numeric vector of B-factor values.

```
library(bio3d)

plot_bfactor <- function(pdb_id, chain = "A", elety = "CA") {
  pdb <- read.pdb(pdb_id)
  pdb_trim <- trim.pdb(pdb, chain = chain, elety = elety)
  bvals <- pdb_trim$atom$b
  plotb3(bvals, sse = pdb_trim, typ = "l", ylab = "Bfactor")
  return(bvals)
}

plot_bfactor("4AKE")
```

Note: Accessing on-line PDB file



##	[1]	29.02	18.44	16.20	19.67	20.26	20.55	17.05	22.13	26.71	33.05
##	[11]	30.66	32.73	25.61	33.19	41.03	24.09	16.18	19.14	29.19	14.79
##	[21]	19.63	28.54	27.49	32.56	17.13	15.50	6.98	24.07	24.00	23.94
##	[31]	30.70	24.70	32.84	34.60	33.01	44.60	50.74	57.32	47.04	67.13
##	[41]	81.04	75.20	59.68	55.63	45.12	39.04	44.31	38.21	43.70	44.19
##	[51]	47.00	48.67	41.54	50.22	45.07	49.77	52.04	44.82	39.75	35.79
##	[61]	38.92	37.93	27.18	26.86	27.53	31.16	27.08	23.03	28.12	24.78
##	[71]	24.22	18.69	40.67	38.08	55.26	46.29	26.25	37.14	27.50	16.86
##	[81]	27.76	19.27	22.22	26.70	25.52	21.22	15.90	15.84	22.44	19.61
##	[91]	21.23	21.79	17.64	22.19	22.73	16.80	23.25	35.95	24.42	20.96
##	[101]	20.00	25.99	24.39	17.19	12.16	17.35	24.97	14.08	22.01	22.26
##	[111]	22.78	27.47	30.49	32.02	20.90	27.03	23.84	44.37	42.47	33.48
##	[121]	44.56	56.67	60.18	66.62	59.95	70.81	88.63	100.11	86.60	85.80
##	[131]	77.48	68.13	52.66	45.34	52.43	60.90	62.64	72.19	66.75	58.73
##	[141]	74.57	79.29	79.53	76.58	66.40	64.76	70.48	74.84	70.11	74.82
##	[151]	78.61	78.24	66.70	66.10	67.01	72.28	80.64	68.54	43.23	51.24
##	[161]	45.72	61.60	45.61	42.57	41.03	41.02	33.34	19.48	34.38	33.11
##	[171]	25.48	29.68	40.71	32.91	24.41	19.20	15.43	19.93	20.66	12.72
##	[181]	21.40	18.21	26.68	34.50	25.77	26.52	36.85	31.05	39.84	48.03
##	[191]	23.04	29.57	23.00	23.80	26.59	25.49	23.25	19.89	32.37	30.97
##	[201]	42.16	29.64	29.69	33.15	26.38	23.17	29.35	32.80	25.92	38.01
##	[211]	45.95	44.26	44.35	70.26						

Example output

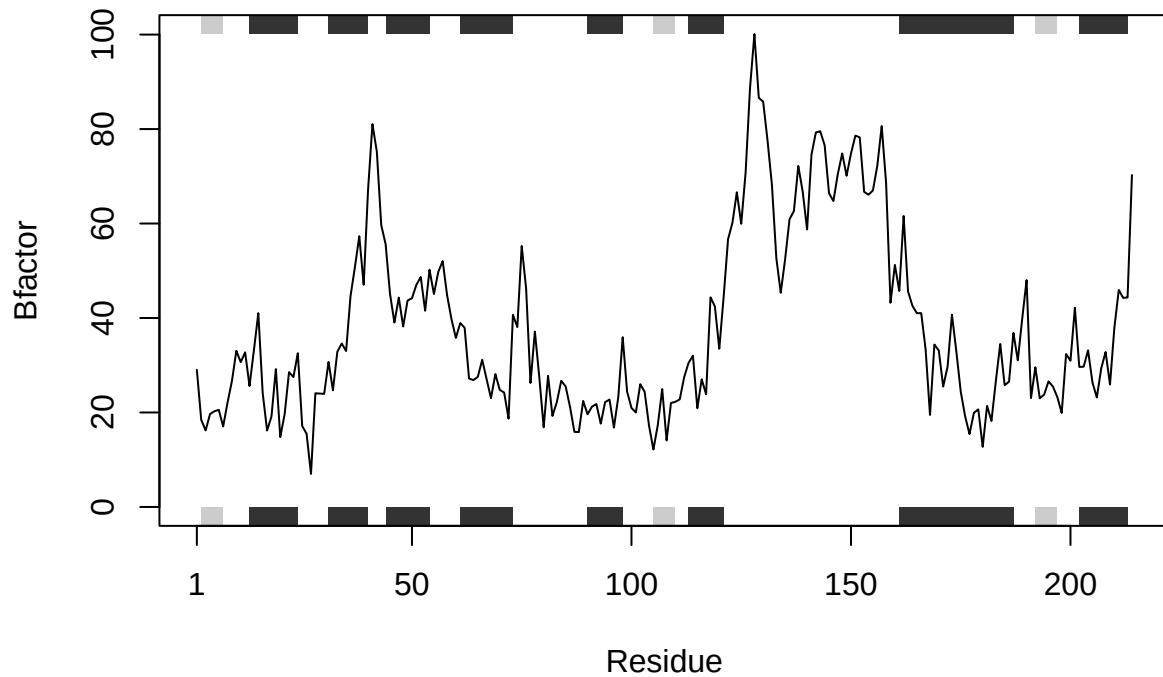
```
s1.b <- plot_bfactor("4AKE")
```

```
## Note: Accessing on-line PDB file
```

```
## Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
```

```
## /var/folders/cg/gk0005qx1gncp164h4t9cg_m0000gn/T//Rtmp9p35CD/4AKE.pdb exists.
```

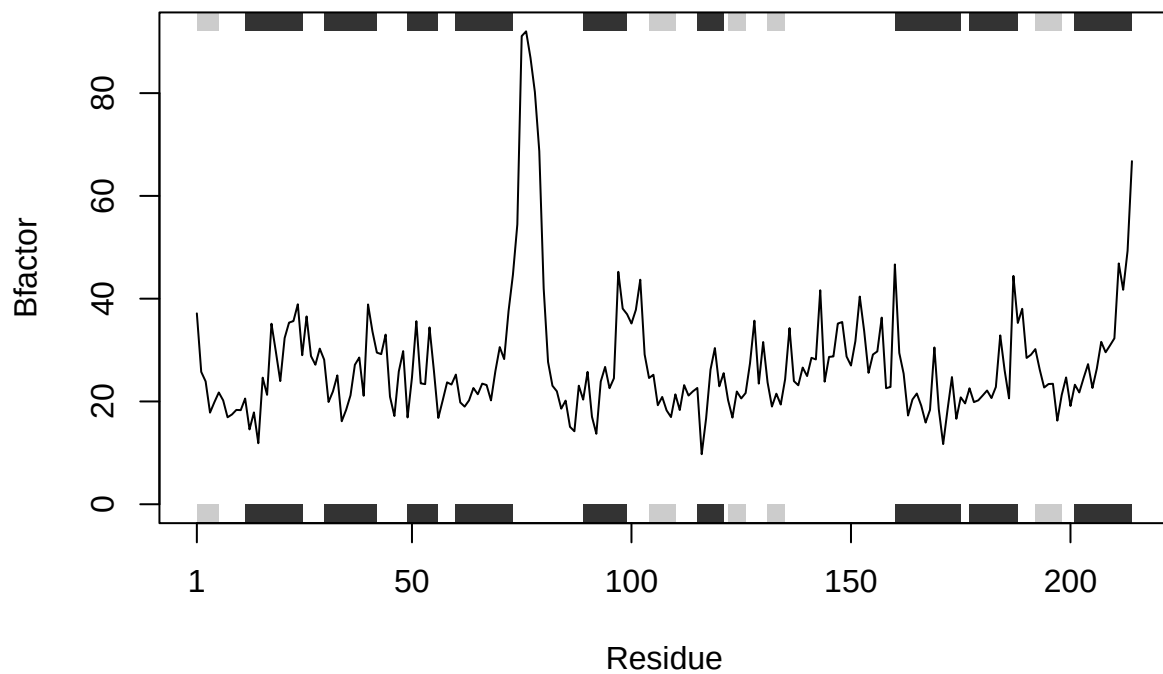
```
## Skipping download
```



```
s2.b <- plot_bfactor("1AKE")
```

```
## Note: Accessing on-line PDB file
```

```
## PDB has ALT records, taking A only, rm.alt=TRUE
```



```
s3.b <- plot_bfactor("1E4Y")
```

```
## Note: Accessing on-line PDB file
```

